

1. S5 Appendix. Additional conceptual figures

1.1. S5.1. Biological motivation for a slow accessibility variable

The figure below situates M_b as a phenomenological compression of slow branch-local processes: spine-neck geometry, actin remodeling, dynamic microtubule entry, local translation, transport readiness, and mitochondrial support. None of these processes is required to map one-to-one onto M_b . Rather, they collectively motivate the class of slow branch-local access constraints that M_b represents.

The figure is provided for conceptual orientation. It does not represent a direct biological measurement, a validated model of any specific molecule, or a claim that cytoskeletal structures directly store memory content. The main article’s claim rests on simulator discrimination, not on the biological sketch.

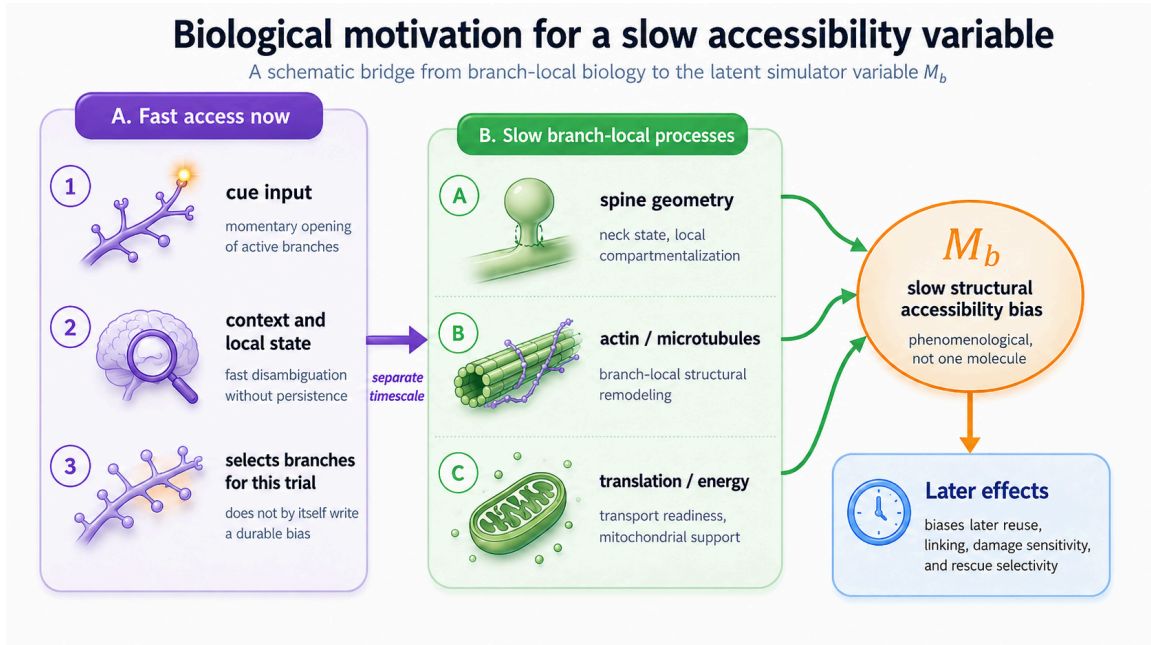


Figure S5.1: **Biological motivation for the slow accessibility variable M_b .** Fast branch opening is driven by cue input, contextual gating, and local dendritic/spine state (left side). Slow branch-local processes — spine geometry, actin dynamics, dynamic microtubule entry, local translation, transport readiness, and metabolic support — evolve on longer timescales and are compressed into the latent variable M_b (right side). M_b is a phenomenological variable: it is biologically motivated and operationally defined, but it should not be read as a direct molecule count or as proof of a cytoskeletal memory code. Candidate processes include spine-neck compartmentalization [1], [2], [3], actin-dependent spine remodeling [4], [5], dynamic microtubule entry into spines [6], [7], [8], [9], [10], branch-local translation [11], [12], [13], and mitochondrial/metabolic support [14], [15], [16].

1.2. S5.2. Structural impedance gating at the spine neck

Microtubule-associated remodeling is one illustrative candidate contributor to structural accessibility [6], [7], [8], [10], [17]. In the schematic below, a low-support structural state is depicted as higher effective impedance at the spine neck, limiting voltage spread from dendrite into spine [1], [2], [3], [18]. A high-support structural state is depicted as lower effective

impedance, permitting signal propagation. This contrast illustrates how slow branch-local structural changes could gate signal flow independently of fast synaptic input strength. The fast/slow access factorization in the model is motivated by this class of slow-timescale process; M_b compresses any combination of them.

This figure is illustrative. The model does not require MT dynamics specifically; M_b may represent any combination of slow branch-local processes.

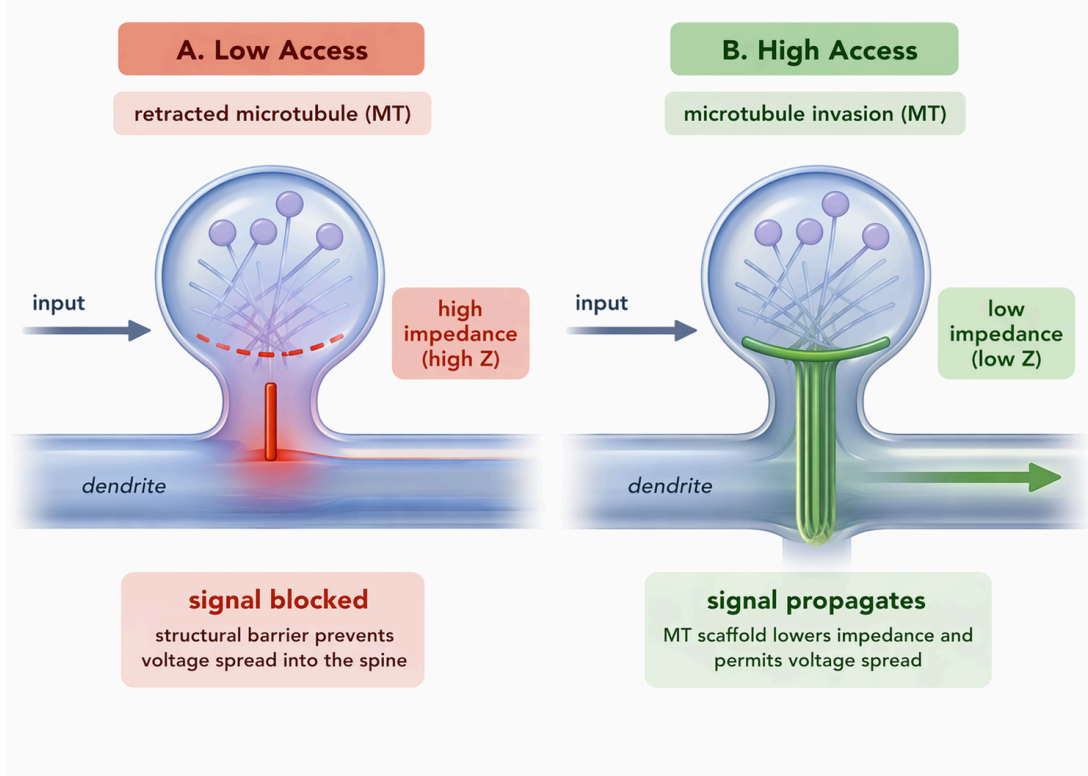


Figure S5.2: **Structural impedance gating at the spine neck (schematic).** (A) Low-support state: the spine-neck structural configuration is schematized as higher effective impedance, limiting voltage spread from dendrite into spine. (B) High-support state: a structurally permissive configuration is schematized as lower effective impedance, permitting signal propagation. Microtubule-associated remodeling is one illustrative candidate contributor; the schematic does not assert that MT dynamics are the unique or dominant mechanism. M_b compresses this and related slow-timescale branch-local processes into a single latent variable.

1.3. S5.3. Minimal biological formalization: variable structure and equations

This figure provides a visual companion to the S1 mathematical appendix. It shows the fast-state variables (x_b, s_t, A_b), the slow-state variables (M_b, E_b, P_b), the memory readout (R_μ), and the two key equations: the fast factorization $A_b = A_f \cdot A_s$ and the slow update $\dot{M}_b$. Readers who find the S1 state-space notation dense may find this visual layout a more accessible entry point.

The figure uses the general biological notation from S1; the main text uses the executable simulator form where $P_{b(t)}W(t)$ makes the consolidation window explicit. See S1 §Correspondence between general and executable forms for the alignment.

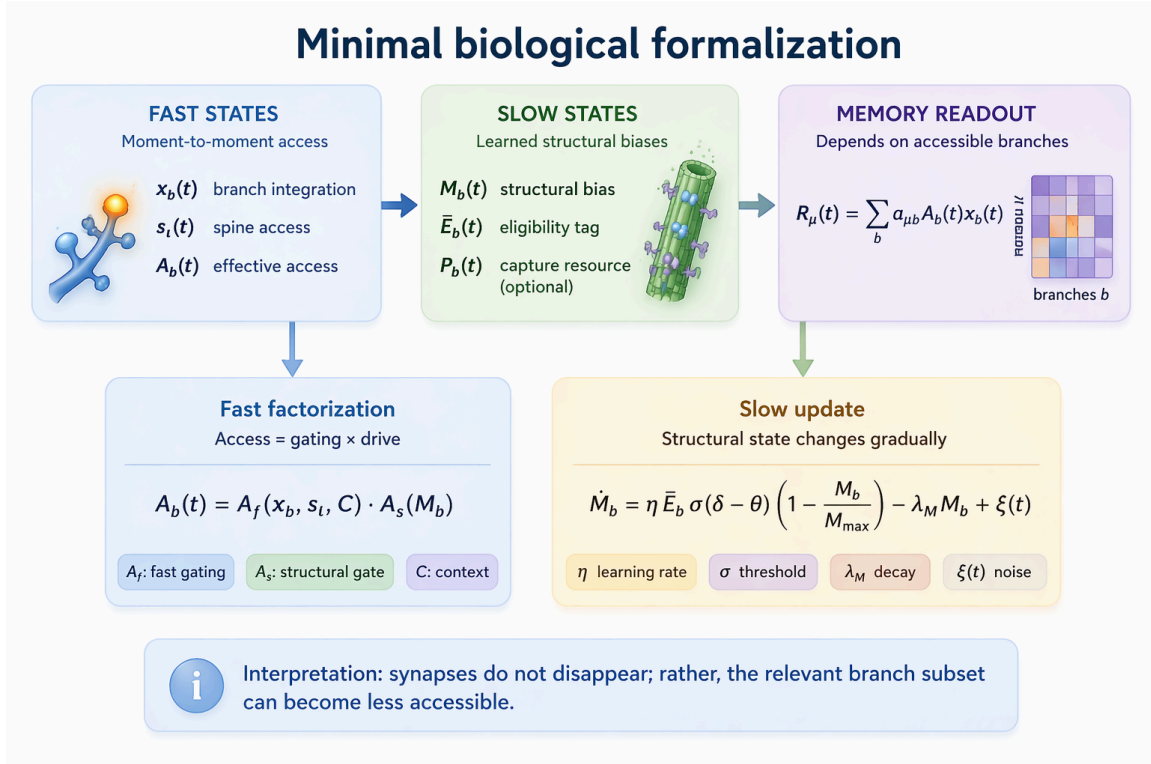


Figure S5.3: **Minimal biological formalization: variable structure and update equations.** Left: fast states (x_b branch integration, s_t spine access, A_b effective access). Centre: slow states (M_b structural bias, $\bar{E}_b$ eligibility tag, P_b capture resource). Right: memory readout $R_\mu = \sum_b a_{\mu b} A_b x_b$. Bottom left: fast factorization $A_b = A_{f(x_b, s_t, C)} \cdot A_{s(M_b)}$. Bottom right: slow update $\dot{M}_b = \eta \bar{E}_b \sigma(\delta - \theta) \left(1 - \frac{M_b}{M_{\max}}\right) - \lambda_M M_b + \xi(t)$. Interpretation note: synapses do not disappear; rather, the relevant branch subset can become less accessible.

1.4. S5.4. Memory linking and contextual retrieval via shared branches

This figure illustrates the key operative prediction: two episodes encoded close in time that share a dendritic branch segment can become linked, so that retrieving one helps recruit the other. An unrelated episode (Episode C) that does not share a branch remains independent. This is the core testbed for the slow-accessibility hypothesis and provides intuition for the simulator's two-trace overlap motif.

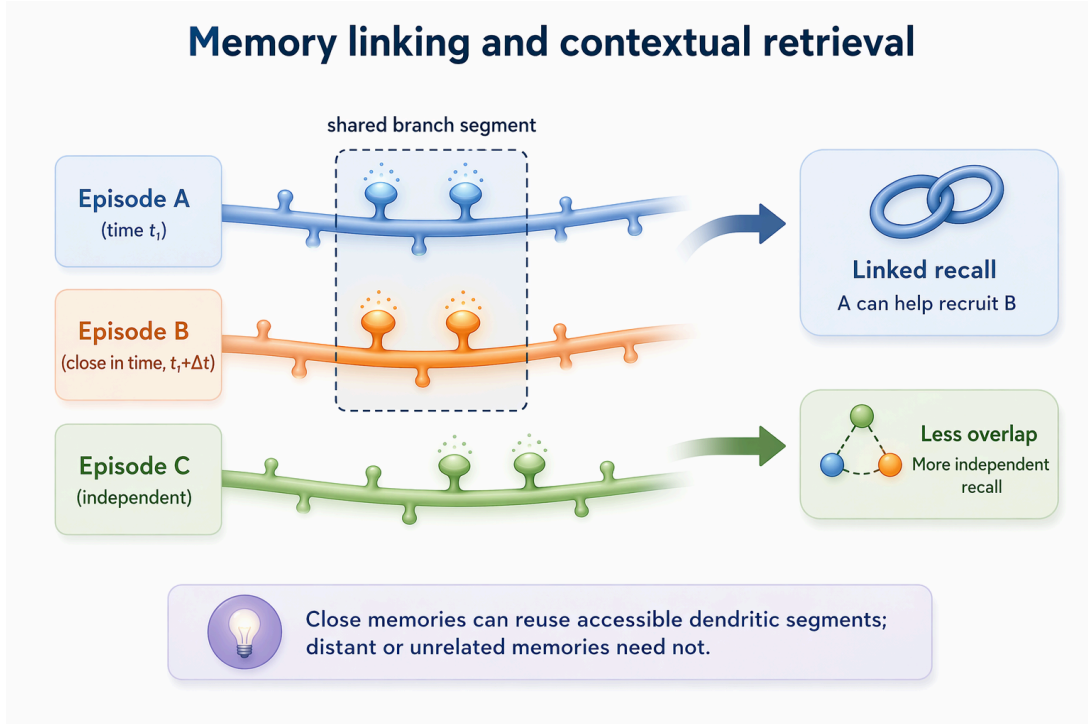


Figure S5.4: **Memory linking and contextual retrieval via shared dendritic branches.** Episodes A and B (encoded close in time, t_1 and $t_1 + \Delta t$) share a branch segment (dashed box). Replay-dependent slow structural writing raises M_b on the shared branch, increasing the linking score $L_{\mu\nu}$; retrieval of A can recruit B (linked recall, upper right). Episode C (independent, no shared branch) remains unlinked (lower right). This schematic motivates the two-trace overlap motif used throughout the simulator experiments and the generalized motif tests in E021–E021R. Memory linking via shared dendritic structure is supported by work on engram neurons, contextual linking, and hippocampal memory allocation [19], [20], [21], [22], [23], [24], [25].

1.5. S5.5. Evidence ladder and claim boundary

The staircase below summarizes the primary simulator-first evidence chain from traceable dynamics to motif boundaries. It also makes the retained claim boundary explicit: supplementary open-data analyses are exploratory bridges, whereas direct biological validation and any unique molecular code claim remain outside the support provided by this paper. Readers unfamiliar with the paper’s overall architecture may find this figure useful as an orientation before reading the Results.

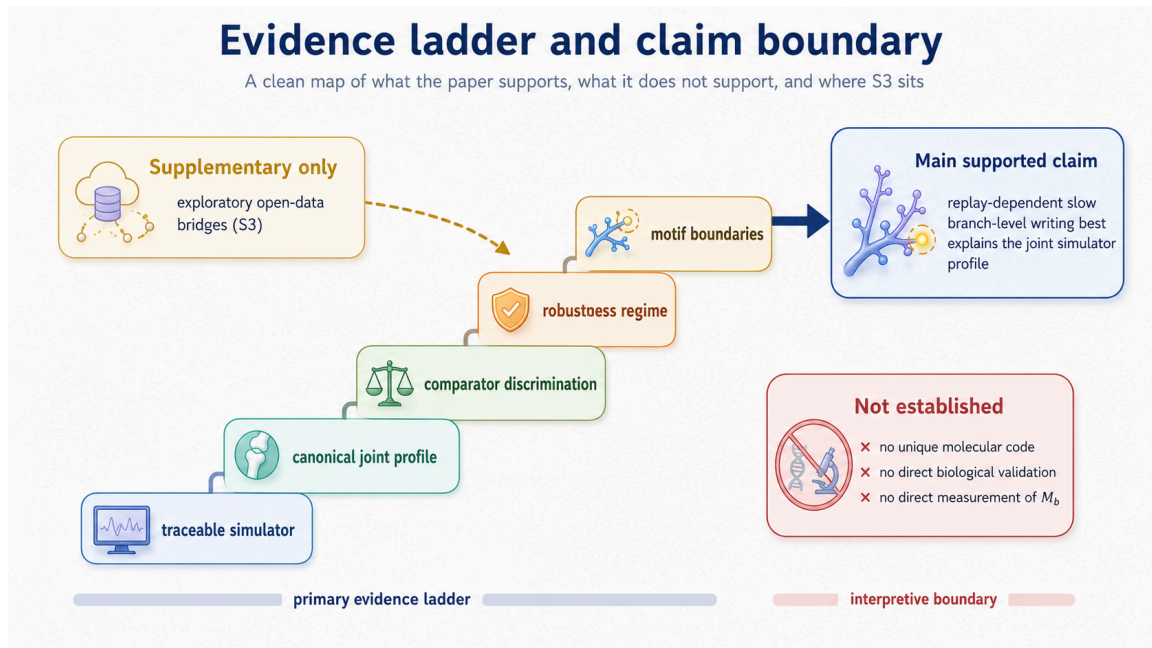


Figure S5.5: **Evidence ladder and claim boundary.** The staircase summarizes the simulator-first evidence chain from traceable branch dynamics (E017) through canonical signatures (E018), robustness (E019–E020), motif generalization (E021–E021R), hard comparators (E022), and replay audit (E022R). The right side marks what the paper does *not* support: direct biological validation of M_b , a unique molecular memory code, or a validated DANDI/open-data claim. The supplementary open-data analyses (S3 Appendix) are exploratory bridges, not primary evidence.

1.6. S5.6. Overlap damage and targeted rescue logic

This figure summarizes the mechanistic logic behind SIG-D and SIG-E. Two traces share an overlap branch that is selectively strengthened by replay-linked slow writing. Focal damage to the shared branch disproportionately reduces linking relative to single-trace recall (SIG-D). Overlap-targeted rescue restores linking more effectively than generic rescue applied to non-overlap branches (SIG-E).

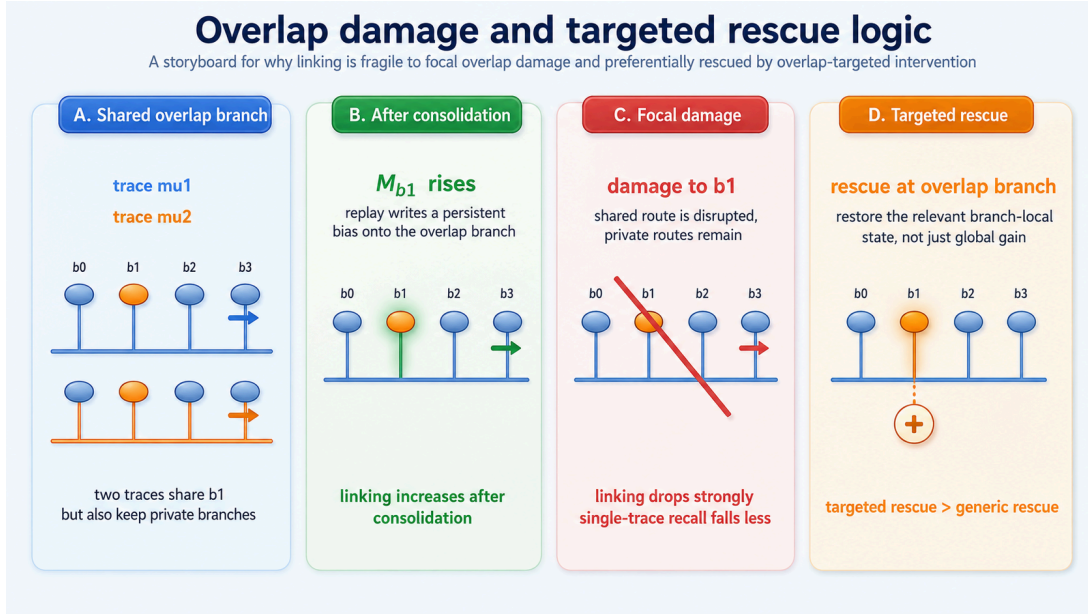


Figure S5.6: **Overlap damage and targeted rescue logic.** Two traces (A, B) share an overlap branch whose slow structural accessibility M_b has been raised by replay-dependent consolidation. Focal damage to the overlap branch selectively reduces the linking score L_{AB} while largely preserving single-trace recall support R_A and R_B (SIG-D). Targeted rescue — which restores eligibility on the overlap branch — recovers linking more than non-targeted generic rescue (SIG-E). This logic motivates the two-tier rescue protocol used in all comparator evaluations.

1.7. S5.7. Comparator fairness and model-discrimination logic

This figure makes explicit the two-level evaluation design used for all comparators. An alternative model is not rejected merely because it lacks M_b . It is first evaluated on output-level metrics (linking gain, damage sensitivity, recovery index) that are computable regardless of mechanism. It is then evaluated on structural mechanistic signatures that require M_b -like branch-specific writing. Only when both levels fail does the alternative fall outside the joint profile.

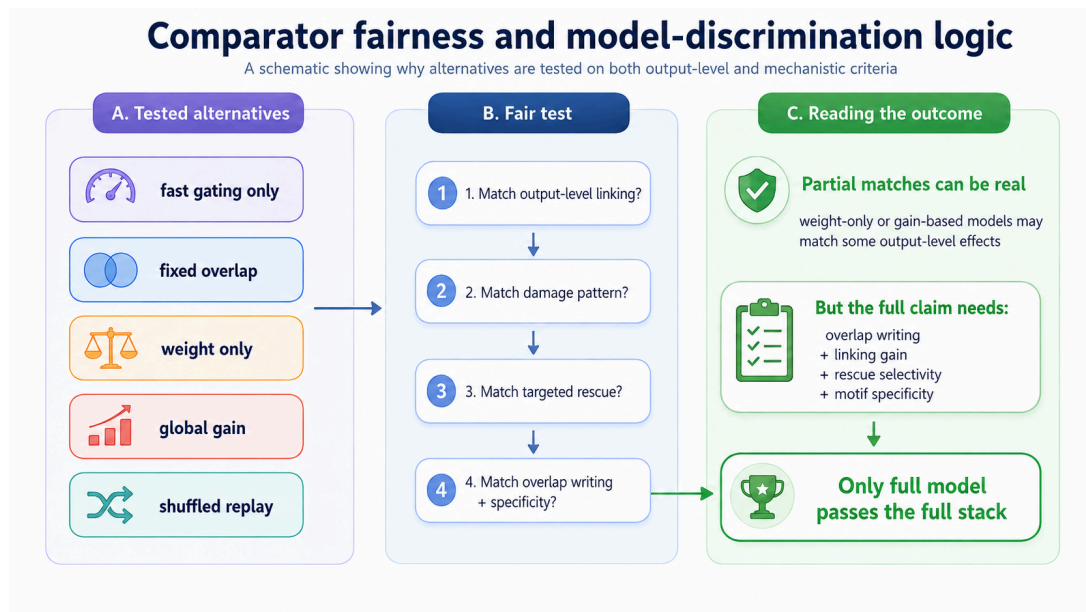


Figure S5.7: **Comparator fairness and model-discrimination logic.** Alternatives are evaluated first on output-level linking and damage patterns (left column: B1, B3, B5 metrics) and then on whether they reproduce overlap-branch structural writing, rescue selectivity, and cross-scale specificity (right column: SIG-A, SIG-B, SIG-E, gSIG-A). Comparators are not dismissed merely for lacking M_b ; they are rejected when they fail the full mechanistic and output-level joint profile. This design prevents circular evaluation and directly addresses the comparator-fairness objection.

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